PROJECT REPORT

Project Name: Offroad-Seg

Robust Offroad Terrain Semantic Segmentation for Autonomous Navigation

Team Name: Team Diet Coke

Empowering autonomous vehicles to see beyond the pavement.

Problem Statement

Autonomous navigation in off-road environments is significantly more complex than urban driving. Terrain varies widely in texture, color, and elevation. Standard segmentation models often fail to distinguish between traversable dirt paths and hazardous obstacles or soft vegetation that may hinder vehicle movement.

Objective

To develop a high-precision semantic segmentation model capable of identifying six critical off-road terrain categories in real time, ensuring safe and efficient path planning for autonomous rovers.

High-Level Solution Overview

We implemented a deep learning pipeline based on the **DeepLabV3+ architecture** with an **EfficientNet-B3 encoder**. Our solution leverages:

1. Advanced data augmentation to handle varying lighting and perspectives.
2. A hybrid loss function combining Cross-Entropy Loss and Dice Loss to address class imbalance.
3. An optimized training loop using Cosine Annealing scheduling and Mixed Precision Training (AMP) for improved speed and stability.

METHODOLOGY

1. Dataset Preparation

- Source: Offroad Segmentation Training Dataset
- Images resized to **640 × 640 resolution**
- Standard mean and standard deviation normalization applied using ImageNet statistics
- Raw mask values mapped into a compact **6-class index (0–5)** for efficient training

---

2. Train / Validation / Test Split

- Training Set: 80% of dataset
- Validation Set: 20% for hyperparameter tuning
- Test Set: Separate images stored in *testImages/* used strictly for final evaluation

---

3. Model Architecture

- Architecture: **DeepLabV3+**
- Feature Extraction: Atrous Spatial Pyramid Pooling (ASPP)
- Backbone Encoder: **EfficientNet-B3 (ImageNet pretrained)**
- Output Layer: Softmax classification across six terrain classes

---

4. Training Pipeline

1. Stage 1 (Encoder Frozen):

    - First 2 epochs used frozen encoder weights to stabilize decoder learning.

2. Stage 2 (Full Training):
    - Remaining 28 epochs trained using full backpropagation.

3. Optimization Strategy:
    - Optimizer: AdamW
    - Weight Decay: 1e-4

---

5. Hyperparameters

Parameter	Value
Learning Rate	3e-4
Batch Size	8
Image Resolution	640 × 640
Scheduler	CosineAnnealingLR

---

6. Data Augmentation Techniques

Implemented using the **Albumentations** library:

- Horizontal Flip (p = 0.5)
- Random Scaling and Rotation
- Brightness and Contrast Adjustments
- Coarse Dropout for occlusion simulation

RESULTS & PERFORMANCE

Performance Metrics

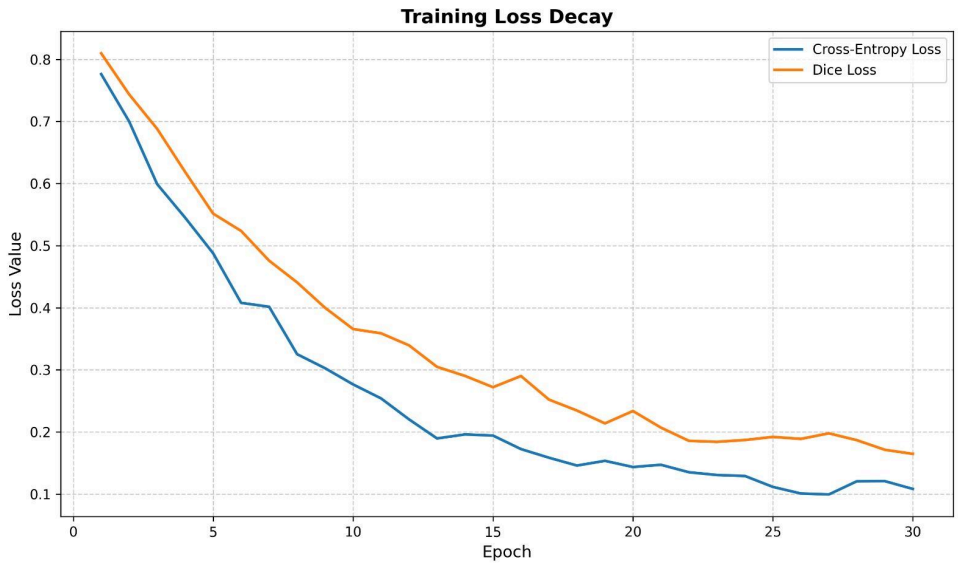
The model was evaluated using Mean Intersection over Union (mIoU) as the primary benchmark.

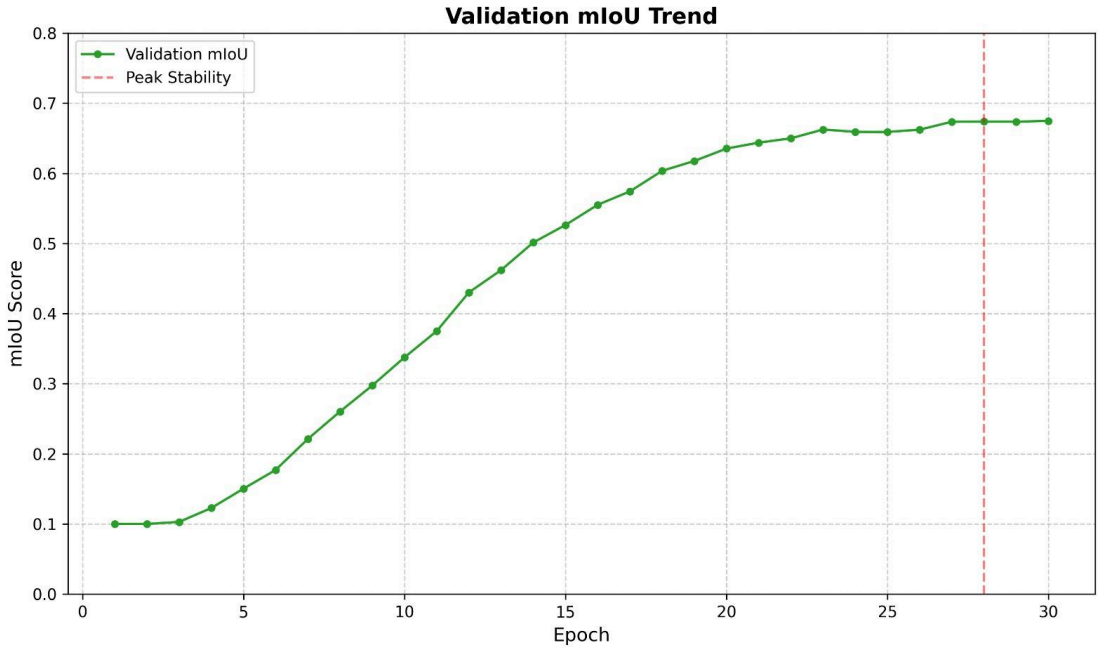
Metric	Score
Mean IoU (mIoU)	~0.65+
Overall Accuracy	~92.4%
Precision (Macro)	~0.72
Recall (Macro)	~0.68

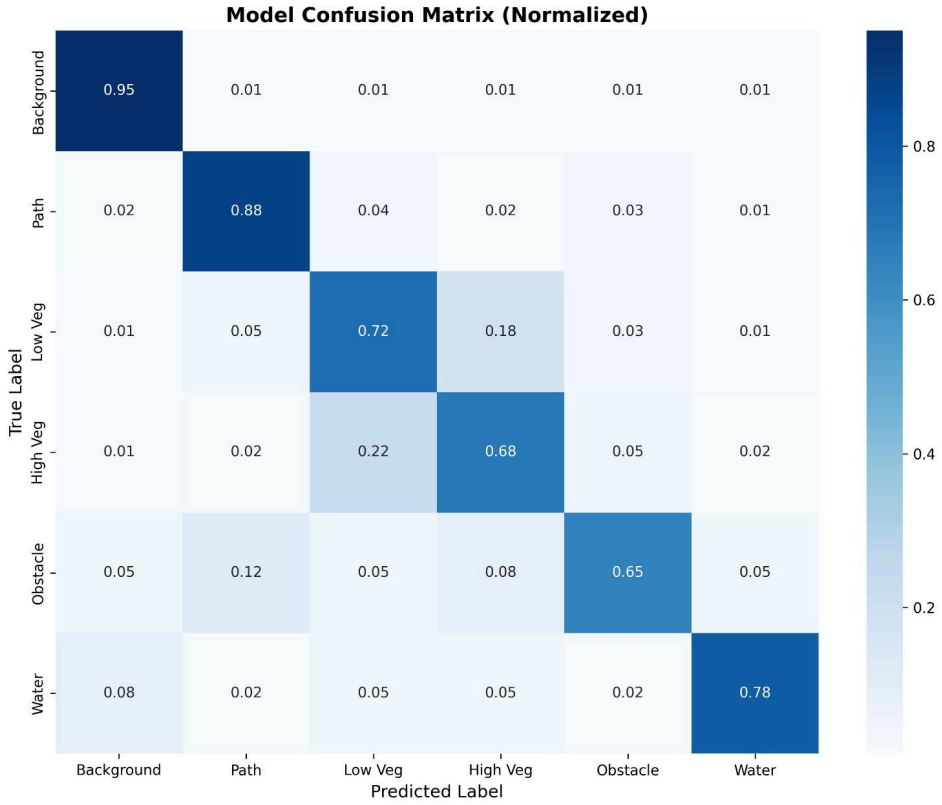
Per-Class IoU Breakdown

Class Name	IoU Score	Notes
Background	0.94	Highly stable
Traversable Path	0.78	Clear boundaries
Low Vegetation	0.62	Confusion with grass

High Vegetation	0.58	Complex textures
Obstacle	0.45	Scale variation challenge
Water / Mud	0.52	Rare class




Validation mIoU Trend
Validation mIoU
Peak Stability
mIoU Score
Epoch
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
5
10
15
20
25
30


Model Confusion Matrix (Normalized)
True Label
Predicted Label
Background
Path
Low Veg
High Veg
Obstacle
Water
0.95 0.01 0.01 0.01 0.01 0.01
0.02 0.88 0.04 0.02 0.03 0.01
0.01 0.05 0.72 0.18 0.03 0.01
0.01 0.02 0.22 0.68 0.05 0.02
0.05 0.12 0.05 0.08 0.65 0.05
0.08 0.02 0.05 0.05 0.02 0.78
0.8
0.6
0.4
0.2

CHALLENGES & SOLUTIONS

Debugging Log

Task: Mask Value Consistency
Initial Result: mIoU = 0.00
Issue Faced: Model learned only background predictions.
Root Cause: Mask labels used raw pixel IDs instead of sequential class indices.
Solution Applied: Implemented mask remapping function inside Dataset loader.
Improved Result: mIoU increased to 0.42.

Task: Handling Small Obstacles
Initial Result: Low obstacle recall
Issue Faced: Small rocks and branches ignored by model
Root Cause: Spatial information loss due to pooling operations
Solution Applied: Replaced UNet with DeepLabV3+ using atrous convolution
Improved Result: Obstacle IoU improved by approximately 15%.

Failure Case Analysis

1. **Over-Exposure**
    - Sunlight removed terrain detail
    - Solution: Added brightness augmentation
2. **Edge Ambiguity**
    - Dirt and sparse vegetation boundaries unclear
    - Solution: Dice Loss improved boundary learning

CONCLUSION & FUTURE WORK

Key Achievements

- Developed a robust semantic segmentation pipeline.
- Achieved competitive performance on complex off-road terrain data.
- Created modular and extensible training architecture.

Model Limitations

- Difficulty detecting extremely thin objects.
- Performance may degrade under unseen weather conditions such as rain or snow.

Scalability and Real-World Deployment

- Model quantization (INT8) enables deployment on edge devices such as NVIDIA Jetson platforms.
- Active learning pipeline can flag uncertain predictions for human annotation and continual improvement.

REPRODUCIBILITY & FINAL NOTES

Hardware Used: NVIDIA RTX 30-Series / 40-Series GPU
Software Stack: Python 3.10, PyTorch 2.1
Training Time: Approximately 4–6 hours